

Exercise 2:

Exercise 2: Aggregate functions

1. COUNT()

- SELECT COUNT(id) AS total_employees
FROM Employees;

Outcome:

total_employees
10

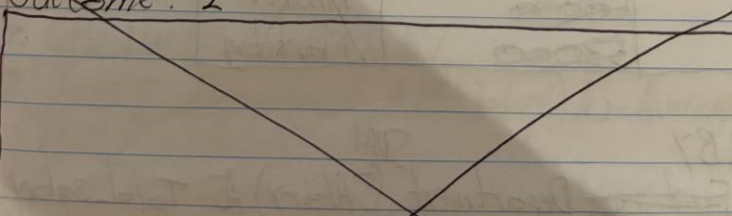
2. SUM()

- SELECT SUM(salary) AS total_salary, Department
FROM Employees
WHERE Department = 'IT';

3. AVG()

- SELECT AVG(salary) AS Average_salary, Department
FROM Employees
WHERE Department = 'HR';

Outcome: 2



Outcome 2:

Total - Salary	Department
22000	IT

Outcome 3:

Average - Salary	Department
49500	HR

49500 HR

4. MIN & MAX

- SELECT MIN(Salary) AS ^{Lowest}~~Minimum~~ Salary, MAX(Salary) AS ^{Highest}~~Highest~~ Salary, Department

FROM Employees

~~GROUP BY~~ Department;

Outcome:

Lowest Salary	Highest Salary	Department
50000 48000 ✓	62000 ✓	IT
59000	57000	HR
52000	60000	Finance
	53000	Marketing

5. GROUP BY ^{SUM}
- SELECT ~~FROM~~ Department, (Salary) AS TotalSalary
FROM Employees
GROUP BY Department;

Outcomes

Department	Total Salary
IT	220000
HR	91000
Finance	11000 119000
Marketing	105000

6. GROUP BY & COUNT()
- SELECT ~~id~~ COUNT(id) AS Number_of_emp
FROM Employees
GROUP BY City;

Outcome:

Num of emp	City
2	New York
3	Chicago
2	Los Angeles
2	San Francisco
1	Houston

7. GROUP BY & ORDER BY
7. SELECT Department, AVG(Salary) AS Average
FROM Employees
GROUP BY Department
ORDER BY Salary DESC;

Outcome 7:

Department	Average salary
IT	55000
HR	49500
Finance	59500
Marketing	52500

8. HAVING CLAUSE
→ SELECT Department, SUM(Salary) AS Total-S
FROM Employees_db
GROUP BY Department
HAVING Total-Salary-paid > 100000;

Outcome:

Department	Total salary Paid
IT	220000
Finance	119000
Marketing	105000

9. SELECT first_name, last_name, city
FROM

9. SELECT city, COUNT(id) AS number-of-employees
FROM employees_db
GROUP BY 1 city
HAVING number-of-employees > 1
ORDER BY number-of-employees DESC;

City	Number-of-employees
New York	2
Chicago	3
Los Angeles	2
San Francisco	2

10. SELECT Department, AVG(Salary) AS Average-Salary
FROM Employees-
GROUP BY Department
HAVING MAX(Average-Salary) > ?

11-03-2024 : SQL JOINS & UNIONS

Employees

1

1001

Salary